

# Mate Competition Manual Report

## Overview:

The MATE ROV Competition is held every year to put to test the creative and critical thinking ability of students in STEM. However, the organizers don't only want the student to learn engineering skills but also learn important real world skills like team work, collaboration, entrepreneurial spirit and more.

The set of skills acquired through participating in this competition is divided into two categories:

1. Employability skills (Orange Signs)
2. ROV specific skills (Green Signs)



## MateROV Philosophy:

“THINK OF YOURSELVES AS ENTREPRENEURS”

From the official Competition manual:

“The 2026 MATE ROV Competition continues to embody this philosophy as well as MATE’s tradition of engaging and empowering participants to do “good for good” for our planet and global community. And with that, you are presented with a request for proposals (RFP), the specifics of which are included below.”

## **Explorer Class Details:**

### **1.0: Competition Overview**

At the World Championship, EXPLORER class companies will undertake four different competition tasks at three different venues. Companies will get ONE attempt at each task.

- TASK #1: Seabed 2023: A Kaleidoscope of Corals in Cold Water
- 2026 EXPLORER CLASS 8
- TASK #2: SmartAtlantic Alliance: Better Information, Better Decisions
- TASK #3: Wind-Powered Offshore Oil Platform: Scalable Solutions for Global Energy Needs
- TASK #4: MATE Floats Under the Ice

World Championship in St. John's, Newfoundland and Labrador allows MATE to offer students the incredible opportunity to operate their technologies in unique, one-of-a-kind venues. The venues include an ice tank (where a thin layer of ice will cover the surface), a wave tank (where waves and wind can be generated), and a flume tank (where a current will be created).

### **1.1 Scoring**

Total 750 points

Scored over 3 distinct categories:

- Product demonstrations
- Engineering & Communication
- Safety

### **1.2 Time**

Each product demonstration includes:

- 5 minutes to set up at the product demonstration station
- 15 minutes to attempt the tasks
- 5 minutes to break down and exit the product demonstration station

(The clock only stops if there arises a situation beyond control else the clock is always ticking)

### 1.3 Evaluation Criteria

- a. Technical documentation
- b. Engineering presentation
- c. Marketing display
- d. Company spec sheet
- e. Corporate responsibility (optional)
- f. Product demonstration
- g. Safety

### 1.4 WEIGHT RESTRICTIONS

Competition officials will use the following chart to award points for weight:

Weight (in air)

| Weight (in air)   |            |
|-------------------|------------|
| < 18 kg           | +10 points |
| 18.01 kg to 25 kg | +5 points  |
| 25.01 kg to 35 kg | +0 points  |

Vehicles greater than **35 kg in weight** will not be allowed to compete in the product demonstration.

### 1.5 ORGANIZATIONAL AND OPERATIONAL EFFECTIVENESS

Companies will receive up to 10 points for Organizational and Operational effectiveness. This includes points for teamwork, collaboration and communication, project management, problem solving and the ability to deal with obstacles, and system design and control.

## 2.0 SAFETY

### 2.1 Jobsite Safety and Environment Analysis

Teams coming to the World Competition must submit Jobsite Safety and Environment Analysis (JSEAs) with technical documentation but as a separate document.

### 2.2 Safety Pre-Inspection

Safety pre-inspection document submissions will include the following:

- Technical documentation
- Company spec sheet
- SID [Electrical, Pneumatic & Hydraulic as utilized]
- Non-ROV device design document (if used)
- Non-ROV device SID (if used)
- Company safety review
- JSEA

The above mentioned things will be presented the day before the competition when the safety pre inspection takes place.

## 2.3 Safety Inspection Protocol

1. Before entering the water for practice or a product demonstration run, the ROV system must go through a safety inspection. Once a company successfully passes inspection, they will turn in their safety inspection sheet to the safety inspector and receive a **Blue PASSED Card** with their company number on it. **Companies must present the Blue PASSED Card to the pool practice/product demonstration coordinator before their vehicles are permitted to enter the water.**

## 3.0 SPECIFICATIONS

The ROV system must meet the following requirements:

- 3.1 Operational
- 3.2 Mechanical
- 3.3 Electrical
- 3.4 Onboard Electrical Power
- 3.5 Power Shutdown
- 3.6 Fluid Power
- 3.7 Control Systems
- 3.8 Command, Control, & Communications (C3)\
- 3.9 MATE Provided Equipment
- 3.10 Laser Safety Rules

## 4.0 COMPETITION RULES

- 4.1 General
- 4.2 Procedural
- 4.3 Design & Safe Considerations

## **5.0 Engineering and Communication**

To emphasize this point, the competition requires the following five engineering and communication components:

- Company spec sheet
- Technical (written) documentation
- Engineering (oral) presentation
- Marketing display
- Corporate Responsibility (OPTIONAL)

## **6. SUBMISSION GUIDELINES AND KEY DEADLINES**

### **6.1 DOCUMENT Submission**

- 6.1.1 technical documentation
- 6.1.2 company spec sheet,
- 6.1.3 SID
- 6.1.4 a fluid power diagram (if fluid power is used
- 6.1.5 a JSEA
- 6.1.6 a company safety review
- 6.1.7 a marketing display
- 6.1.8 corporate responsibility effort document
- 6.1.9 a non-ROV device design document
- 6.1.10 a non-ROV device SID
- 6.1.11 photo of their sonar attachment device

### **6.2 Video Submission**

6.3 Mention of all the deadlines sequentially (Didn't copy paste the entire deadlines but can be found at the end of the competition manual)